

Felica Card API

Development Documentation

V1.0.0

Update Records

Date	Description
2024-07-04	First Edition Release

Contents

- 1 Command return value description
- 2 Error code description
- 3 Operating instructions
- 4 Operation analysis
 - 4.1 Polling
 - 4.2 Read
 - 4.3 Write
- 5 Android API
 - 5.1 Open
 - 5.2 Close
 - 5.3 Card searching
 - 5.4 Read data
 - 5.5 Write data
- 6 Reference documents
- 7 Demo test process

1 Command return value description

Success: no data: success (PS: return value is 0)

With data: relevant command demand data (e.g. read)

Failed: error: (error code)

2 Error code description

Negative error code: (mostly loss of communication between PICC and PCD, poor contact, re check the card)

Error Code	Description
-1	Error in parsing input parameters of single chip microcomputer
-2	APK layer interface input parameter error
-10	APK layer transmission error
-1001	Parameter error
-1002	IO error
-1003	timeout error
-1004	Check error
-1005	framing error
-1006	Interference error
-1007	Protocol error
-1008	Card conflict error
-1009	Send error

3 Operating instructions

In our demo, you can test it as follow:

- 1. Click "open NFC" to open NFC and place the card in the identification area
- 2. Click "polling" to poll the card. If there is a card, you can continue to execute "read" and "write"

4 Operation analysis

Note: the input parameter is hexadecimal, and the return value with data output is hexadecimal

4.1 Polling

FF FF 00 03

Yellow area: System Code

Green area: Request Code

Blue area: Time Slot Number

Return Value

Success: length (1 byte) + response code (1 byte) + IDM (8 bytes) + pad (8 bytes)

Failed: error: (error code)

4.2 Read

01 09 00 01 80 00

Yellow area: Number of Services

Green area: Service Code List

Blue area: Number of Blocks

Purple area: Block List

Return Value

Success: Length (1 byte) + response code (1 byte) + IDM (8 bytes) + Status flag1 (1 byte)+ Status Flag2 (1 byte) + Number of blocks (1 byte) + block data (16 * m bytes (M is the value of number of blocks))

Failed: Error: (error code)

4.3 Write

01 09 00 01 80 00 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F

Yellow area: Number of Services

Green area: Service Code List

Blue area: Number of Blocks

Purple area: Block List

Grey area: Block Data

Return Value

Success: success

Failed: error: (error code)

5 Android API

Package:com.common.apiutil.nfc.Nfc

5.1 Open

```
/**
 * @Function Open the NFC device
 *
 * @Exception CommonException
 * 1.DeviceAlreadyOpenException: The device has been turned on
 * 2.DeviceNotFoundException: The device cannot be opened
```

```

*
* */

public synchronized void open() throws CommonException

```

5.2 Close

```

/**
 * @Function Close the NFC device
 *
 * @Exception CommonException
 *
 * */

public synchronized void close()throws CommonException

```

5.3 Card searching

```

/**
 * @Function Card searching
 *
 * @Parameter
 * 1.cmd: cmd->4 bytes; System code/Reserved/Time slot number
 * 2.cmdLen: cmd length
 *
 * @Return Success, card data; Failed, CommonException
 *
 * */

public synchronized byte[] FelicaPolling(byte[] cmd, int cmdLen) throws
CommonException

```

System code	Reserved	Time slot number
2 bytes	1 byte '00'	1 byte

5.4 Read data

```

/**
 * @Function Read data
 *
 * @Parameter
 * 1.cmd: cmd->determines the number of bytes according to the set parameter;
Number of Services/Service Code List/Number of Blocks/Block List
 * 2.cmdLen: cmd length
 *

```

```

* @Return Success, card data; Failed, CommonException
*
* */

```

```

public synchronized byte[] FelicaRead(byte[] cmd, int cmdLen) throws
CommonException

```

Command	bytes	Multiple	Description
Number of Services	1	n	It depends on the implementation of the Type 3 Tag how many Services can be read simultaneously.
Service Code List	2n		Each Service Code is specified in Little Endian format.
Number of Blocks	1	m	It depends on the implementation of the Type 3 Tag how many blocks can be read simultaneously.
Block List	2m-3m		See Section 5.6.

5.5 Write data

```

/**
* @Function Write data
*
* @Parameter
* 1.cmd: cmd->determines the number of bytes according to the set parameter;
Number of Services/Service Code List/Number of Blocks/Block List/Block Data
* 2.cmdLen: cmd length
*
* @Return Success, card data; Failed, CommonException
*
* */

```

```

public synchronized byte[] FelicaWrite(byte[] cmd, int cmdLen) throws
CommonException

```

Command	bytes	Multiple	Description
Number of Services	1	n	It depends on the implementation of the Type 3 Tag how many Services can be written simultaneously.
Service Code List	2n		Each Service Code is specified in Little Endian format.

Command	bytes	Multiple	Description
Number of Blocks	1	m	It depends on the implementation of the Type 3 Tag how many blocks can be written simultaneously.
Block List	2m-3m		See Section 5.6.
Block Data	16m		

6 Reference documents

Please refer these online document:

<Type 3 Tag Operation Specification.pdf>

<JIS-X-6319-4-2005.pdf>

7 Demo test process

- 1. Open NFC
- 2. Polling
- 3. Read
- 4. Write
- 5. Close NFC